

Autodesk® Scaleform®

Scaleform 4

Scaleform (Scaleform 3) Scaleform 4.0

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1.03

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Autodesk® Scaleform® 4.2

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Autodesk Scaleform

Scaleform 4

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1 Scaleform 4

Scaleform v4
Autodesk
SDK

UI SDK
,
ActionScript 3.0
,Scaleform

,Scaleform v4
AS 2.0
AS 3.0

AS3,
SDK
,Scaleform v4

- w/ 2x+
–
2.5D
- Scaleform 4
Advance
Display
- ,AS3
AS2
Scaleform 4
- AS3
(Code Base),
,
- /
–
,AS3
Flash
- AS 2.0
,
AS 3.0
(Runtime Exception)
- AS 2.0
,
AS 3.0
,
,
- ActionScript 3.0
,
(Method Closure)

Scaleform 4
,
Scaleform 4
Scaleform 4.0
4.2
,Scaleform 4
Scaleform 3.x
,
,

Scaleform
Scaleform 4
,
Scaleform C++
:
(NameSpace)
,

2

2.1

```
namespace  
namespace Scaleform  
{  
    class Foo  
    {  
        ...  
    };  
}
```

- 1) `Scaleform::Foo var;`
- 2) `using namespace Scaleform;`
`Foo var;`

Scaleform, C++
Scaleform, "Scaleform"

- Scaleform, "using"
- Scaleform "SF" (SF = Scaleform)
(Header);

- SF , :“Render” “Sound” “GFx”
- , ,“Win32” “PS3”,
 , “Scaleform::Render::D3D9” ,
 “SF::D3D9::Render”

2.2 Scaleform

Scaleform , Scaleform
 (Public)

Scaleform , , ,
 Scaleform

Scaleform 4

- Scaleform
 - o GFx
 - AMP
 - XML
 - o Render
 - Math2D
 - D3D9
 - GL
 - PS3
 - ...
 - o Text
 - o Sound
 - o Alg
 - o UTF8Util

3

Scaleform , Scaleform

, ,

- “Scaleform” Src
- “Src/Kernel”

- Src
 - o GFx
 - AMP
 - AS2
 - AS3
 - ...
 - o Kernel
 - HeapMH
 - HeapPT
 - o Render
 - PS3
 - GL
 - D3D9
 - ...
 - o Sound
 - o XML

Src , cpp (Public) ,

4 Includes

Scaleform 3.x ,Scaleform 4 /Src , /Include
.cpp

Scaleform , ' (convenience) ,Gfx.h
;Gfx_Kernel.h Kernel , Include

,
include

```
#include "Kernel/SF_File.h"  
#include "Gfx/Gfx_Player.h"  
#include "Gfx/Gfx_Loader.h"  
#include "Gfx/Gfx_Log.h"
```

```
#include "Gfx_Kernel.h"  
#include "Gfx.h"
```

, ,

"PublicHeader : GroupName"

,GroupName , Gfx Kernel Render AMP ... none GroupName()
Gfx, Gfx.h , GroupName Kernel, Gfx_Kernel.h ,
GroupName 'none', -
, ,

5

```

Scaleform 4 API , , Scaleform ,
    Scaleform , Scaleform ( Scaleform 3.x SDK
), Scaleform 4 API
Scaleform include Scaleform 4
    Scaleform 3.x Scaleform 4
Scaleform Scaleform
    " Scaleform 4 (Getting Started with Scaleform 4.2) "

    Scaleform Scaleform 4
        Scaleform 4
        Scaleform

Scaleform 3.x
Scaleform 4
Scaleform 4 Scaleform
    " Scaleform 4 (Scaleform 4Renderer Threading Guide) "

,Scaleform 4 (Include/GFxCompat.h),
Scaleform 3.x Scaleform 4 typedefs enums defines
Scaleform 3.X Scaleform 4
    Scaleform 4
    Scaleform 4
Apps/Demos/GFxPlayerTiny/GFxPlayerTinyD3D9Compat.cpp

GFXCompat.h Scaleform 4 ,
, GFX Loader Scaleform 4 :FontProvider AS2Support /
AS3Support ,

// Set Font Provider:
Ptr<FontProviderWin32> fontProvider = *new FontProviderWin32(::GetDC(0));
loader.SetFontProvider(fontProvider);

// Add AS2 Support:
Ptr<ASSupport> pAS2Support = *new GFx::AS2Support();
loader.SetAS2Support(pAS2Support);

// Add AS3 Support:
Ptr<ASSupport> pASSupport = *new GFx::AS3Support();

```

```
loader.SetAS3Support(pASSupport);
```

```
, Scaleform 3.x Scaleform 4
GfxCompat.h
```

5.1

```
Scaleform 3.x 4 , , Scaleform 4
```

```
#include "Gfx_Kernel.h"
#include "Gfx.h"
#include "Gfx_Renderer_D3D9.h" // or whatever platform you need
```

```
(Gfx.h) AS3 AS3
"Gfx/AS3/AS3_Global.h" AS3
Scaleform LITE
```

AS3_Global.h and Obj\AS3_Obj_Global.xxx

```
:AS3_Global.h Obj\AS3_Obj_global.xxx
AS3_Obj_Global.xxx " "ActionScript 3 swf
"(script) , GlobalObjectCPP, C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++ , ,
ClassRegistrationTable,
AS3_Global.h

ClassRegistrationTable include
ClassRegistrationTable( ) AS3_Global.h
, ( , ),
, AS3 VM ,
, " " ,
```

5.2

Scaleform 3.X, GFC, SF_4, Scaleform 4, GFX_4

Scaleform 3.X	Scaleform 4
GFC_FX_VERSION_STRING	GFX_VERSION_STRING
GUNUSED	SF_UNUSED
GFC_64BIT_POINTERS	SF_64BIT_POINTERS
...	...

5.3 Namespace

Scaleform 4 namespace, (Scaleform::Gfx::Event) 'using namespace', using, cpp

```
namespace SF = Scaleform;
using namespace Scaleform;
using namespace Render;
using namespace Gfx;
```

5.4

Scaleform 4, ,

Scaleform 3.X	Scaleform 4
UInt	unsigned
SInt	int
Float	float
...	

Scaleform 4, Scaleform, GfxEvent, Scaleform 4, Gfx::Event, Scaleform 3.x, Scaleform 4

Scaleform 3.X	Scaleform 4	Scaleform 4
		'using namespace ...'

GFxSystem	Scaleform::GFx::System	System
GPtr	Scaleform::Ptr	Ptr
GFxMovieDef	Scaleform::GFx::MovieDef	MovieDef
GFxMovieView	Scaleform::GFx::Movie	Movie
GRendererD3D9	Scaleform::Render::D3D9::Renderer	Render::D3D9::Renderer
GFxRenderConfig	Scaleform::GFx::RenderConfig	RenderConfig
GString	Scaleform::String	String
GFxFSCCommandHandler	Scaleform::GFx::FSCCommandHandler	FSCCommandHandler
GFxLog	Scaleform::GFx::Log	Log
GFxImageCreator	Scaleform::GFx::ImageCreator	ImageCreator
GFxKey	Scaleform::GFx::Key	Key
GFxKeyEvent	Scaleform::GFx::KeyEvent	KeyEvent
GFxCharEvent	Scaleform::GFx::CharEvent	CharEvent
GFxEvent	Scaleform::GFx::Event	Event
GMatrix2D	Scaleform::Render::Matrix2x4	Matrix2x4
GMatrix3D	Scaleform::Render::Matrix3x4	Matrix3x4
GMatrix3D	Scaleform::Render::Matrix4x4	Matrix4x4
GRect	Scaleform::Render::Rect	Rect
...	...	...

6 Scaleform 4.0 4.2

Scaleform 4.2 4.0 API Scaleform 4.1
API 4.0

6.1

SF 4.2 PCRE Perl (Perl Compatible
Regular Expressions, PCRE), AS3 3rdParty ,
Scaleform 4.2 PCRE AS3,
PCRE(Include/GFxConfig.h SF_ENABLE_PCRE),
PCRE

6.2

Scaleform 4.2 , , Scaleform Scaleform
(Build System)
Render() Sound() SF 4.2
Render Scaleform
X360/D3D9/D3D1x , InitHAL()
• HALConfig_DynamicShaderCompile
Visual Studio , : Render , ShaderData.xml,
Custom Build Step

6.3

Scaleform 4.2 -Scaleform 4.2
40 SF 4.0 Scaleform 4.0
gfxlicense.txt sf_license.txt
gfx.lib
,sf_license.txt Scaleform AMP Scaleform
Scaleform IME sf_video_license.txt
sf_ime_license.txt

Scaleform